JUXTAPOSING A KOSZUL-REGULAR SEQUENCE WITH A SEQUENCE THAT IS KOSZUL-REGULAR MODULO THE IDEAL IT GENERATES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Koszul regularity of a sequence and the identification of the Koszul complex as a finite free resolution of the quotient. assumed in the statement, source lines 7639-7646. Supplies both hypotheses and the goal.
- The Koszul complex of a juxtaposed sequence is the total complex of the tensor product of the two Koszul complexes. the author cites Lemma lemma-join-sequences-koszul-complex at proof line 7660. Performs the first identification of the chain.
- A double complex whose rows are resolutions computes the homology of the total complex. the author cites Homology, Lemma homology-lemma-double-complex-gives-resolution at proof lines 7662-7666. Justifies the first quasi-isomorphism in the displayed chain.

Source reference. Original lemma 0669: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7928–7928. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0664: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7923–7923. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0133: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1421–1421. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring. Let I be an ideal generated by a Koszul-regular sequence $f_1, \ldots, f_n$ in A. Let $g_1, \ldots, g_m \in A$ be elements whose images $\overline{g}_1, \ldots, \overline{g}_m$ form a Koszul-regular sequence in A/I. Then $f_1, \ldots, f_n, g_1, \ldots, g_m$ is a Koszul-regular sequence in A.*

Proof. Our assumptions say that $K_\bullet(A, f_1, \ldots, f_n)$ is a finite free resolution of A/I and $K_\bullet(A/I, \overline{g}_1, \ldots, \overline{g}_m)$ is a finite free resolution of $A/(f_i, g_j)$ over A/I. Then

$$\begin{aligned} K_\bullet(A, f_1, \ldots, f_n, g_1, \ldots, g_m) &= \mathrm{Tot}(K_\bullet(A, f_1, \ldots, f_n) \otimes_A K_\bullet(A, g_1, \ldots, g_m)) \\ &\cong A/I \otimes_A K_\bullet(A, g_1, \ldots, g_m) \\ &= K_\bullet(A/I, \overline{g}_1, \ldots, \overline{g}_m) \\ &\cong A/(f_i, g_j) \end{aligned}$$

The first equality by Lemma 0664. The first quasi-isomorphism $\cong$ by (the dual of) Homology, Lemma 0133 as the qth row of the double complex $K_\bullet(A, f_1, \ldots, f_n) \otimes_A K_\bullet(A, g_1, \ldots, g_m)$ is a resolution of $A/I \otimes_A K_q(A, g_1, \ldots, g_m)$. The second equality is clear. The last quasi-isomorphism by assumption. Hence we win. $\square$